IC104 Introductory Quantitative Techniques for Business and Finance

Lecture 2. Differentiation, Derivatives and Single-Variable Optimisation

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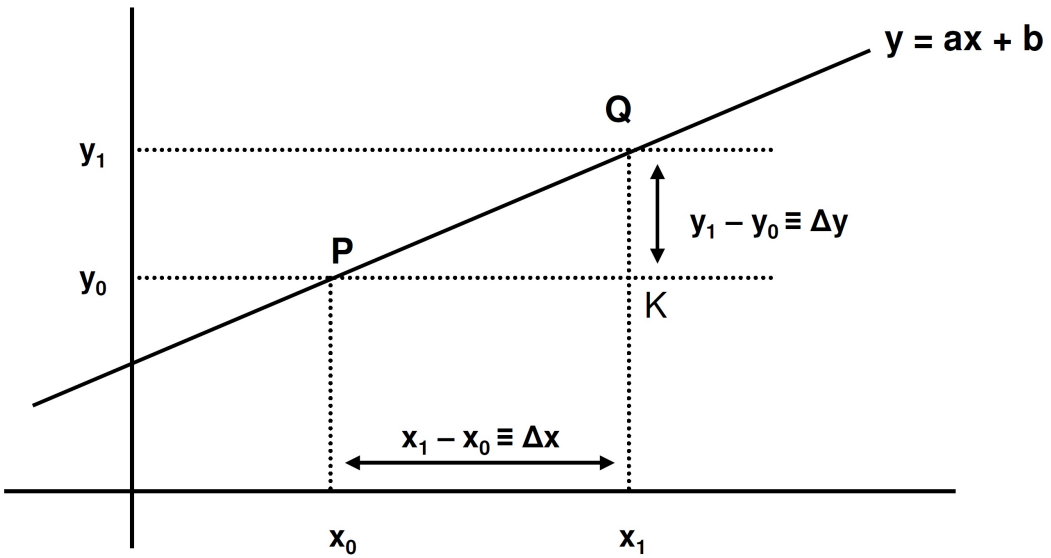
ICMA Centre
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Lecture Plan

- Differentiation(Sydsaeter & Hammond Chapter 6)
- Derivatives in Use (Sydsaeter & Hammond Chapter 7)
- Single-Variable Optimisation (Sydsaeter & Hammond Chapter 8)

Introduction to Differentiation

- Recall the general form of a linear function: $y = ax + b$.
- Assuming that a and b are both positive, the graph of that linear function is:



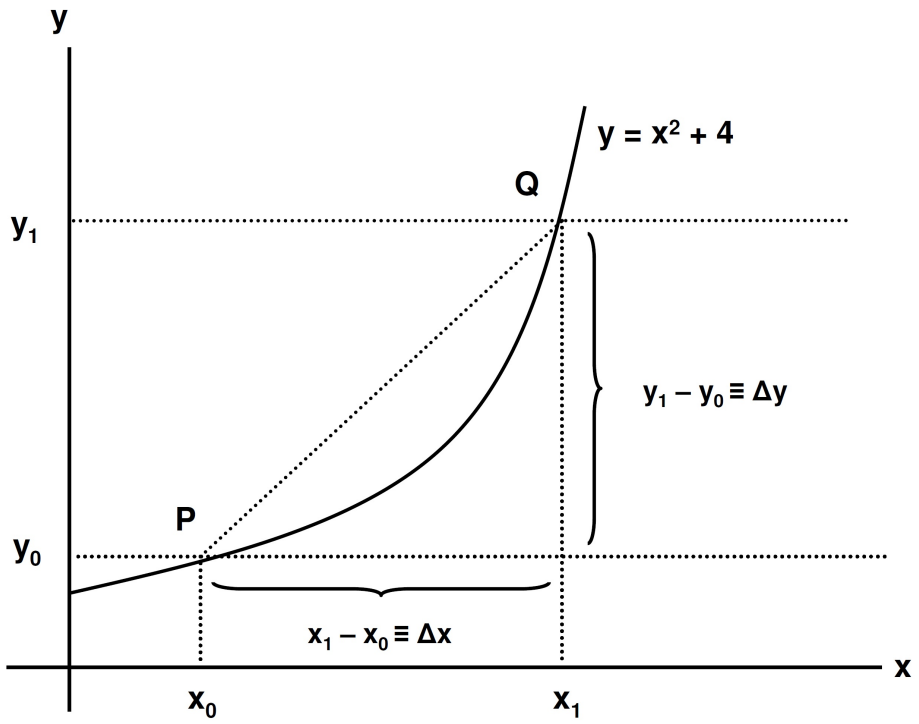
- The **slope** (or **gradient**) of the function is:

$$\frac{QK}{PK} = \frac{y_1 - y_0}{x_1 - x_0} = \frac{\Delta y}{\Delta x} = \frac{\text{change in } y}{\text{change in } x}$$

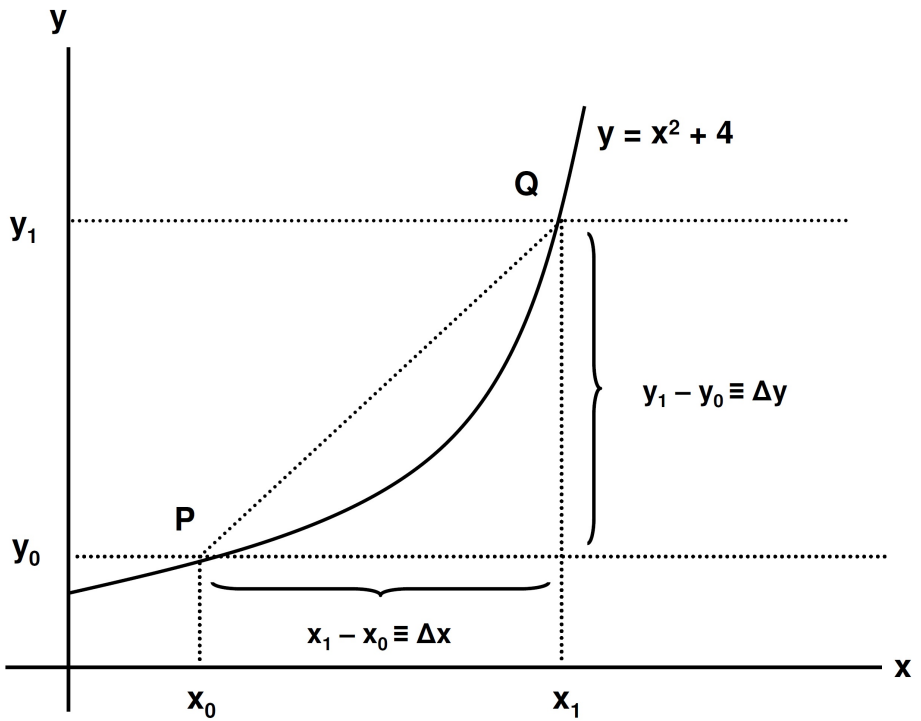
- This ratio gives us the change in y per unit of change in x. We call this the **rate of change** in y.

The Slope of a Curved Line

- Suppose we want to find the slope (gradient) of the curve $y = x^2 + 4$.
- A feature of the slope of the curve is that it is constantly changing. As x increases, and we move up the curve, the slope gets steeper.

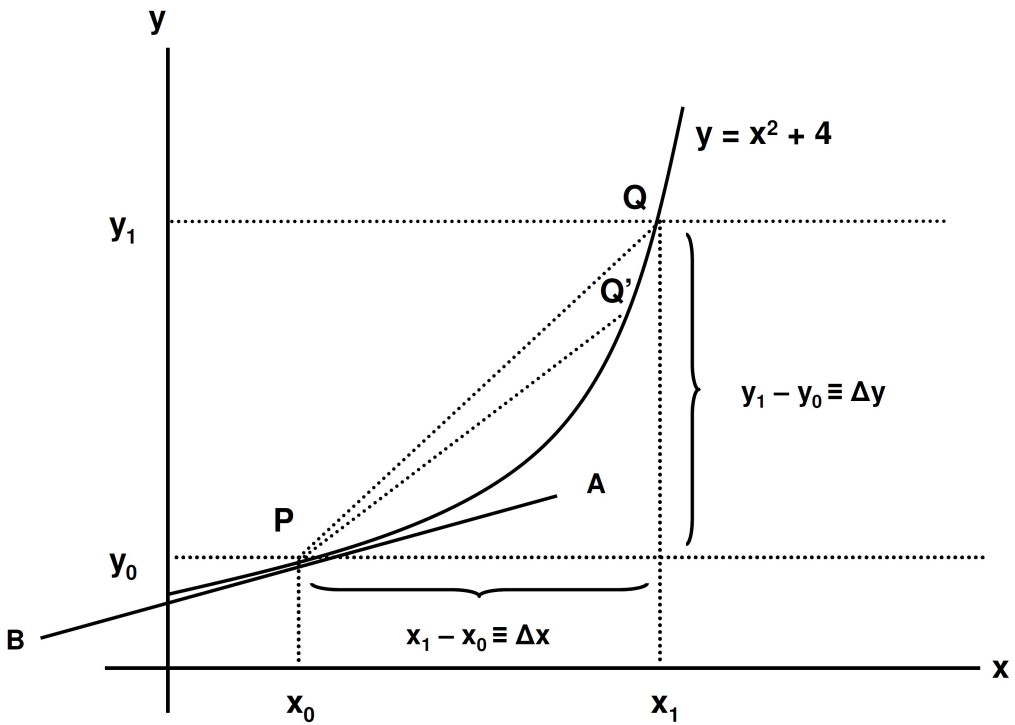


The Slope of a Curved Line



- If we measure the slope as before with a linear function by using $\frac{y_1 - y_0}{x_1 - x_0}$, then we are not really measuring the slope of the curve, but the slope of the straight line joining P and Q.
- If instead we draw a tangent line to the curve, then we can measure the slope of the function at that tangency point. A tangent to a curve is defined as a line which touches the curve only at one point.

The Slope of a Curved Line



- Line AB is a tangent to the curve at P. The slope of the tangent line in P is unique; a line with a larger or smaller slope would touch the curve at another point.
- Next, we're going to move Point Q closer and closer to Point P. As we do so, line PQ will come closer and closer to the tangent line AB.

The Slope of a Curved Line

- From the graph, since P lies on the curve we know that $y_0 = {x_0}^2 + 4$. Also Q lies on the curve, so we know that $y_1 = {x_1}^2 + 4$.
- Therefore, $y_1 - y_0 = [{x_1}^2 + 4] - [{x_0}^2 + 4]$
- By definition, $\Delta x = x_1 - x_0$. Rearrange this to get: $x_1 = x_0 + \Delta x$
- Substitute $x_1 = x_0 + \Delta x$ into $y_1 - y_0$

$$
\begin{aligned}
y_1 - y_0 &= [(x_0 + \Delta x)^2 + 4] - [{x_0}^2 + 4] \\
\Delta y &= [{x_0}^2 + 2x_0\Delta x + \Delta x^2 + 4] - [{x_0}^2 + 4] \\
\Delta y &= 2x_0\Delta x + \Delta x^2 \\
\Rightarrow \frac{\Delta y}{\Delta x} &= \frac{y_1 - y_0}{x_1 - x_0} = \frac{2x_0\Delta x + \Delta x^2}{\Delta x} \\
&= 2x_0 + \Delta x
\end{aligned}
$$

- As Δx approaches zero, the slope of the line PQ approaches the slope of the tangent AB. We write this, $\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = 2x_0$

The Slope of a Curved Line

- In mathematical terms:

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \frac{dy}{dx} = \text{slope of the tangent at } P$$

- Or, $f'(a) = \left.\frac{dy}{dx}\right|_{x=a}$, the slope of the tangent to the curve $y = f(x)$ at the point $(a, f(a))$
- And the derivative of the function f at point a, denoted by $f'(a)$ is:

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

- And the equation for the tangent line to the curve $y = f(x)$ at point $(a, f(a))$ is

$$y - f(a) = f'(a)(x - a)$$

Example

- Compute $f'(a)$ when $f(x) = x^3$
- Solution:
  - $f(a+h) = (a+h)^3 = a^3 + 3a^2h + 3ah^2 + h^3$
  - $f(a+h) - f(a) = a^3 + 3a^2h + 3ah^2 + h^3 - a^3 = 3a^2h + 3ah^2 + h^3$
  - $$\frac{f(a+h) - f(a)}{h} = \frac{3a^2h + 3ah^2 + h^3}{h} = 3a^2 + 3ah + h^2$$
  - As h goes to zero, all the terms containing h disappear.
  - $\lim_{h \to 0} = 3a^2$, so $f'(a) = 3a^2$

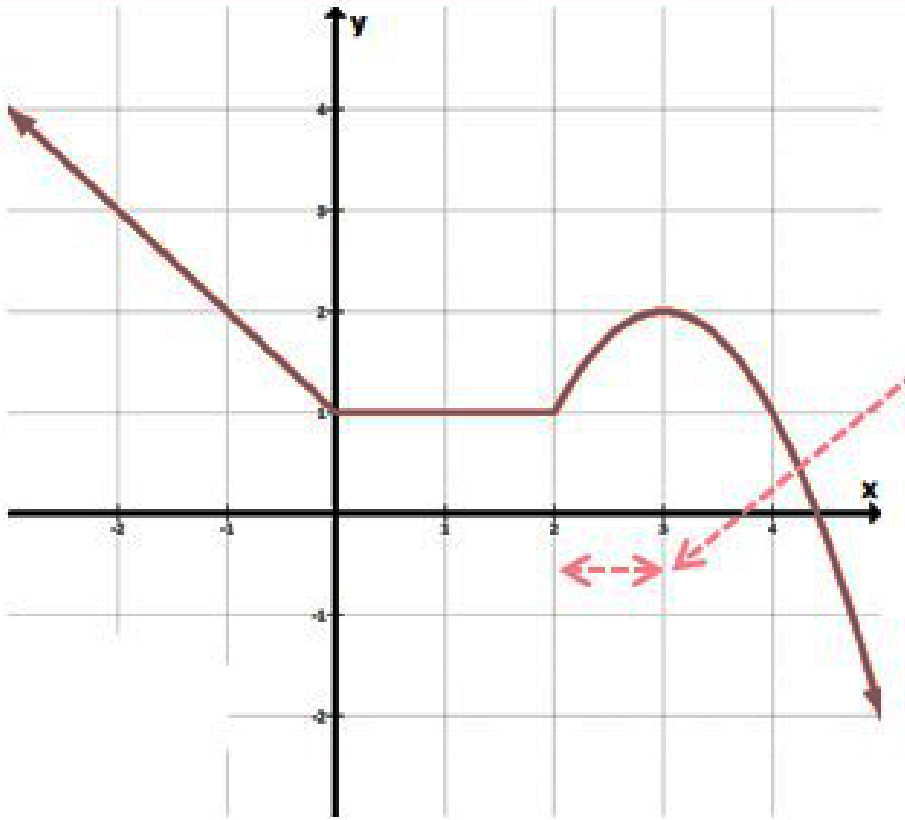
Increasing and Decreasing Functions

- If $f'(x) \geq (>)0$ in the interval $I \Leftrightarrow f$ is increasing (strictly increasing) in I
- - that is, if the value of the derivative is positive here, the graph is sloping upwards.

- If $f'(x) \leq (<)0$ in the interval $I \Leftrightarrow f$ is decreasing (strictly decreasing) in I
- - that is, if the value of the derivative is negative here, the graph is sloping downwards.

- If $f'(x) = 0$ in the interval $I \Leftrightarrow f$ is constant in I
- - that is, if the value of the derivative is zero here, the graph is flat.

Increasing and Decreasing Functions

Graph	Intervals
	***f(x)*** **is increasing:** $(2, 3)$ (Notice this is not a point, but an **interval of the x**) ***f(x)*** **is decreasing:** $(-\infty, 0) \cup (3, \infty)$ ***f(x)*** **is constant:** $(0, 2)$

Differentiation Application: Rate of Change

- Economists interpret the derivative in many important ways:
  - The instantaneous rate of change of f at a is $f'(a)$.
  - The relative rate of change of f at a is $\frac{f'(a)}{f(a)}$.
- Example 1: If $K(t)$ is the capital stock in an economy at time t, $K'(t)$ is called the rate of investment at time t. It is usually denoted by I(t). So $K'(t) = I(t)$.
- Example 2: Let $C(x)$ be the cost of producing x units of product for a firm. The derivative $C'(x)$ at x is called the marginal cost at x, i.e. it is the cost of producing one more unit at that point.

$$C'(x) = \lim_{h \to 0} \frac{C(x+h) - C(x)}{h}$$

Rules for Differentiation

- In the following we state the main rules to find a derivative of a function $f(x)$. The process of finding a derivative is known as differentiation.
- Power rule (a is an arbitrary constant)
  - $f(x) = x^a \Rightarrow f'(x) = ax^{a-1}$
  - Example: $y = x^2 \Rightarrow y' = 2x$
  - Example: $y = x^{-2} \Rightarrow y' = -2x^{-3}$
  - Example: $y = x \Rightarrow y' = 1 \cdot x^0 = 1$
- Multiplicative and additive constants (A is a constant)
  - $f(x) = A \Rightarrow f'(x) = 0$
  - $y = A + f(x) \Rightarrow y' = f'(x)$ (Additive constants disappear)
  - $y = Af(x) \Rightarrow y' = Af'(x)$ (Multiplicative constants are preserved)
  - Example: $y = 3x^4 - 4 \Rightarrow y' = 12x^3$

Rules for Differentiation

- Differentiation of Sums and Differences
  - $F(x) = f(x) \pm g(x) \Rightarrow F'(x) = f'(x) \pm g'(x)$
  - i.e. If $F(x) = f(x) + g(x) \Rightarrow F'(x) = f'(x) + g'(x)$
  - i.e. If $F(x) = f(x) - g(x) \Rightarrow F'(x) = f'(x) - g'(x)$
  - Where functions are added together, we can simply add their derivatives together.
- Example: Profit function $\pi(x) = R(x) - C(x)$, where $R(x) = 3x^2$ and $C(x) = 12x$.
- The marginal profit is $\pi'(x) = R'(x) - C'(x) = (3x^2)' - (12x)' = 6x - 12$

Rules for Differentiation

- Derivative of A Product
  - $F(x) = f(x) \cdot g(x) \Rightarrow F'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$
  - We keep one function the same and take the derivative of the other, and then take the derivative of the first and keep the second the same.
  - Example: $y = (1 - x^2)(2x - 4)$

$$f(x) = (1 - x^2), f'(x) = -2x, g(x) = 2x - 4 \text{ and } g'(x) = 2$$
$$\Rightarrow F'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x) = (-2x) \cdot (2x - 4) + 2(1 - x^2)$$
$$= -4x^2 + 8x + 2 - 2x^2 = -6x^2 + 8x + 2$$

  - In general, $F(x) = f_1(x) \cdot f_2(x) \cdot ... \cdot f_n(x)$:

$$\Rightarrow F'(x) = f_1'(x) \cdot f_2(x) \cdot ... \cdot f_n(x) + f_1(x) \cdot f_2'(x) \cdot ... \cdot f_n(x) + ...$$
$$+ f_1(x) \cdot f_2(x) \cdot ... \cdot f_n'(x)$$

Rules for Differentiation

- The Derivative of A Quotient:
  - $F(x) = \dfrac{f(x)}{g(x)} \Rightarrow F'(x) = \dfrac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{(g(x))^2}$
  - Example $F(x) = \dfrac{2x}{1+x^2}$, $f(x) = 2x$, $g(x) = 1 + x^2$

$$F' = \frac{2 \cdot (1+x^2) - 2x \cdot (2x)}{(1+x^2)^2} = \frac{-2x^2 + 2}{(1+x^2)^2}$$

- Function of a function rule (chain rule): If $y = f(u)$, where $u = g(x)$, the derivative is:
  - $\dfrac{dy}{dx} = \dfrac{dy}{du} \cdot \dfrac{du}{dx}$
  - Example: $y = (2x^2 + 4x)^2$, $y = u^2$, and $u = 2x^2 + 4x$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 2u \cdot (4x + 4) = 2(2x^2 + 4x)(4x + 4)$$

Chain Rule

- What happens if we have one function inside another?
- Suppose f (x) and g(x) are differentiable functions, and the composite function $F(x) = f(g(x))$ is differentiable.
- Then the derivative of this composite function is:

$$F' = f'(g(x))g'(x)$$

- First, differentiate the outer function, keeping the inner function the same, and then multiply by the derivative of the inner function.
- Alternative notation: let $u = g(x)$ and $y = f(u) = f(g(x))$, then

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Chain Rule: Examples

- $y = (x^3 + x^2)^{50}$
- Start by identifying the inner and outer functions:

$$y = u^{50}, \quad u = x^3 + x^2$$

- Next, find the derivative of the inner function:

$$u' = 3x^2 + 2x$$

- Finally, apply the Chain Rule:

$$y' = 50u^{50-1} \cdot u' = 50(x^3 + x^2)^{49}(3x^2 + 2x)$$

Chain Rule: Examples

- $y = \sqrt{x^2 + 1}$
- Start by identifying the inner and outer functions:

$$y = u^{\frac{1}{2}}, \quad u = x^2 + 1$$

- Next, find the derivative of the inner function:

$$u' = 2x$$

- Finally, apply the Chain Rule:

$$y' = \frac{1}{2}u^{\frac{1}{2}-1} \cdot u' = \frac{1}{2}(x^2+1)^{-\frac{1}{2}} \cdot 2x = \frac{x}{\sqrt{x^2+1}}$$

Chain Rule: Examples

- Chain rule and quotient rule together:
- $y = (\frac{x-1}{x+3})^{\frac{1}{3}}$

$$y = u^{\frac{1}{3}}, \quad u = \frac{x-1}{x+3}, \quad u' = \frac{1(x+3)-(x-1)1}{(x+3)^2} = \frac{4}{(x+3)^2}$$

$$y' = \frac{1}{3}u^{\frac{1}{3}-1} \cdot u' = \frac{1}{3}(\frac{x-1}{x+3})^{-\frac{2}{3}} \cdot \frac{4}{(x+3)^2}$$

Higher-Order Derivatives

- $f'(x)$ of a function $f(x)$ is called the first derivative of $f(x)$.
- $f'(x)$ is also a function. If $f'(x)$ is also differentiable, its derivative $(f')'(x) = f''(x)$ is called the second derivative of f.
  - $f(x), f' = \frac{dy}{dx}, f'' = \frac{d^2y}{d^2x}$
- Find f' and f'', when $f(x) = 2x^5 - 3x^3 + 2x$

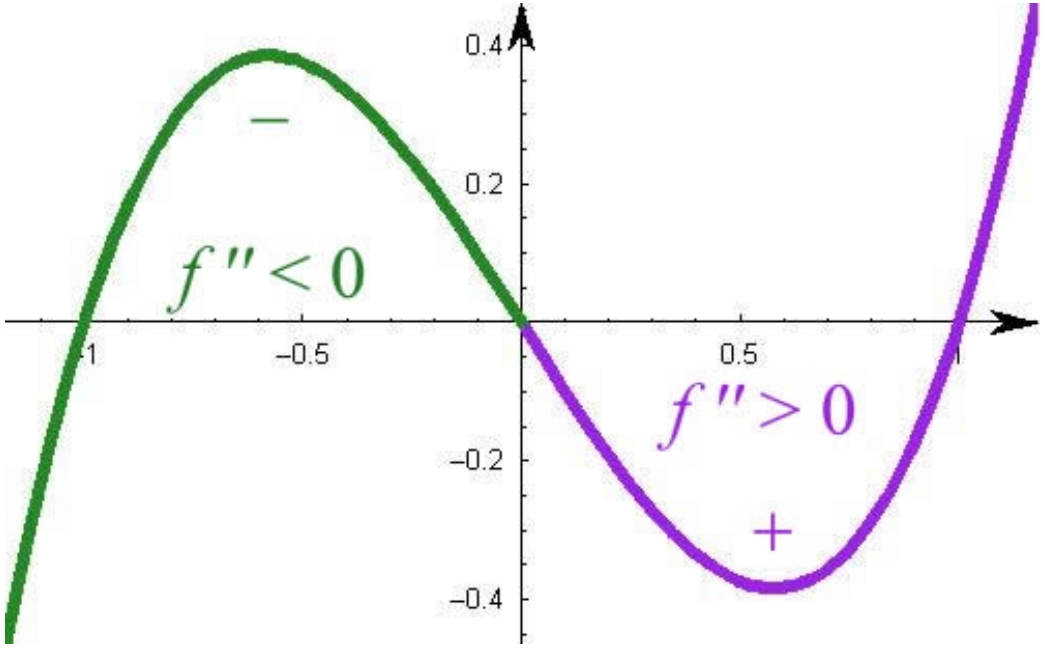
$$f' = 10x^4 - 9x^2 + 2$$
$$f'' = 40x^3 - 18x$$

- There's no reason to stop at differentiating twice. Provided $f^{(n-1)}(x)$ is differentiable, nth-order derivatives are:

$$y^{(n)} = f^{(n)}(x) \text{ or } \frac{d^n y}{d^n x}$$

Convex and Concave Functions

- If $f''(x) > 0$ on interval $I \Leftrightarrow f'$ is increasing on I
- If $f''(x) < 0$ on interval $I \Leftrightarrow f'$ is decreasing on I
- f is **convex** on $I \Leftrightarrow f''(x) \geq 0$ for all x in I
- f is **concave** on $I \Leftrightarrow f''(x) \leq 0$ for all x in I



Exponential and Logarithmic Functions

- $y = e^x \Rightarrow y' = e^x$
- Remember from the previous section: the derivative of the exponential function is itself. This is only the case for this function.
- $y = a^x \Rightarrow y = (e^{\ln a})^x = e^{(\ln a)x} \Rightarrow y' = a^x \ln a$
- $y = \ln x \Rightarrow y' = \frac{1}{x}$
- $y = \ln h(x) \Rightarrow y' = \frac{h'(x)}{h(x)}$
- Examples:

$$y = x^p e^{ax} \Rightarrow y' = px^{p-1}e^{ax} + x^p ae^{ax} = x^{p-1}e^{ax}(p + ax)$$

$$y = x2^{3x} \Rightarrow y' = 1 \cdot 2^{3x} + x \cdot 3 \cdot 2^{3x} \ln 2 = 2^{3x}(1 + 3xln2)$$

$$y = \ln(4 - x^2) \Rightarrow y' = \frac{-2x}{4 - x^2} = \frac{2x}{x^2 - 4}$$

Implicit Differentiation

- How to differentiate functions defined implicitly by an equation such as $g(x, y) = c$
- The Method of Implicit Differentiation:
  - Differentiate each side of $g(x, y(x)) = c$ with respect to x, considering y as a function of x (Chain Rule).
  - Solve the resulting equation for y'
- Example $y^3 + 3x^2y = 13$

$$3y^2y' + 6xy + 3x^2y' = 0$$

$$\Rightarrow y' = \frac{-6xy}{3x^2 + 3y^2} = \frac{-2xy}{x^2 + y^2}$$

Economic Example

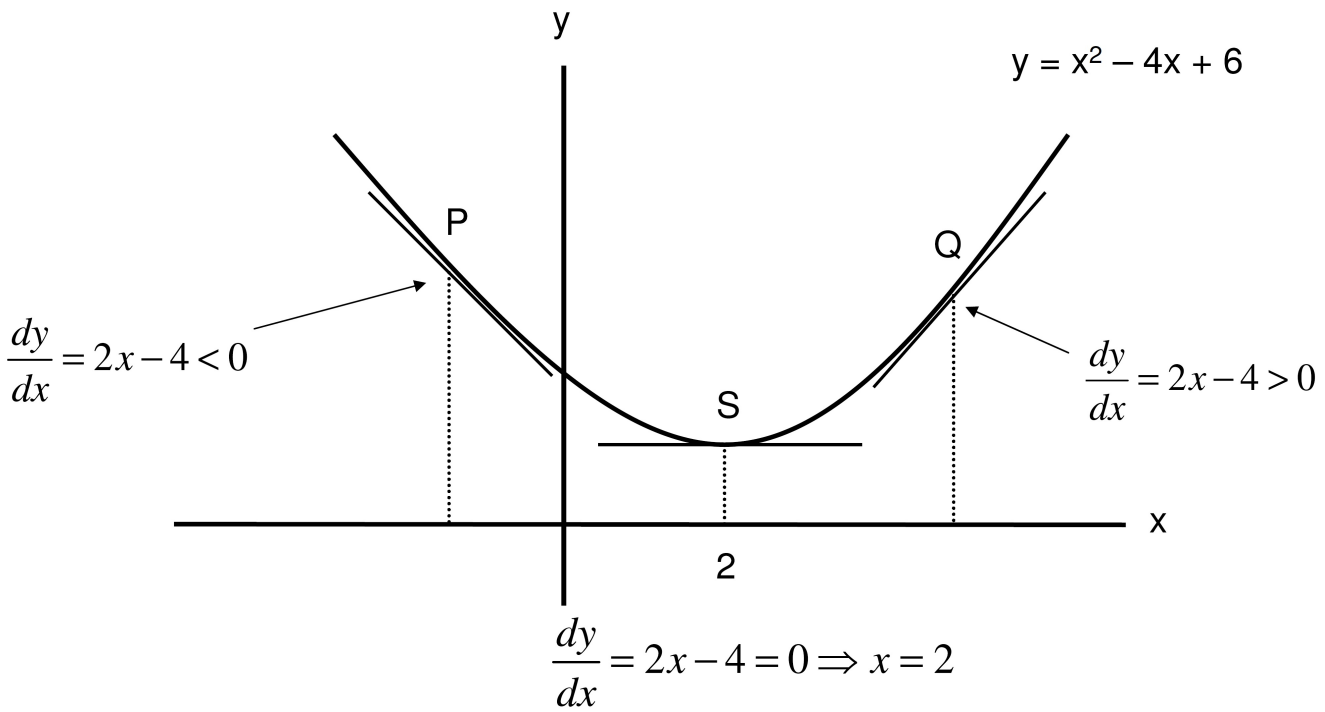
- Suppose the supply function is $S = AP^{-\alpha}r^{-\beta}$. The supply of a commodity is a decreasing function in its price P and the interest rate r. How does the equilibrium price react to an increase in the interest rate?
  - We need to find $\frac{dP}{dr}$
  - Differentiating the supply function:
    $A(-\alpha P^{-\alpha-1})\frac{dP}{dr} \cdot r^{-\beta} + AP^{-\alpha}(-\beta r^{-\beta-1}) = 0$
  - Solving for $\frac{dP}{dr}$: $\frac{dP}{dr} = \frac{AP^{-\alpha}\beta r^{-\beta-1}}{A(-\alpha P^{-\alpha-1})r^{-\beta}}$
  - $= \frac{-1}{\alpha} \cdot \frac{A}{A} \cdot \frac{P^{-\alpha}}{P^{-\alpha-1}} \cdot \frac{r^{-\beta-1}}{r^{-\beta}}$
  - $= -\frac{P\beta}{\alpha r} < 0$

Maximum and Minimum Values

- How do we find maximum and minimum points?
- Very often in Finance and Economics, we need to find the point of maximum profitability.
- The key is this: at a maximum and minimum point, the slope of the function is equal to zero.
- This means that we need to differentiate the function, set the derivative equal to zero, and solve for the maximum or minimum point.
- We need to check whether the point is a maximum point, a minimum points or a point of inflection by looking at the Second Derivative.

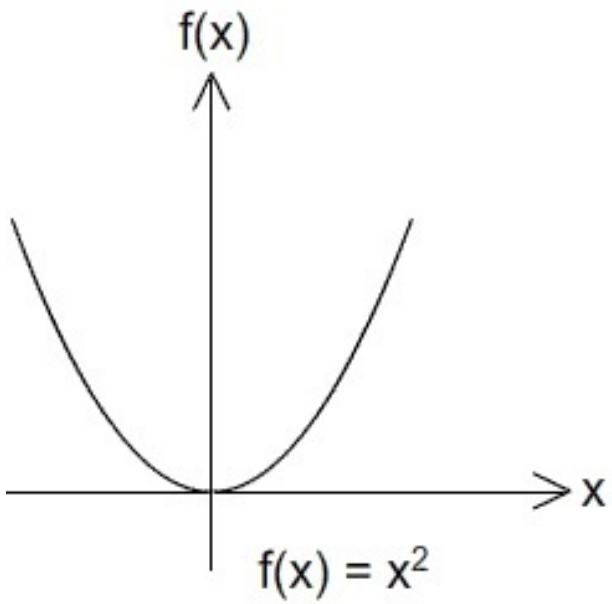
Maximum and Minimum Values

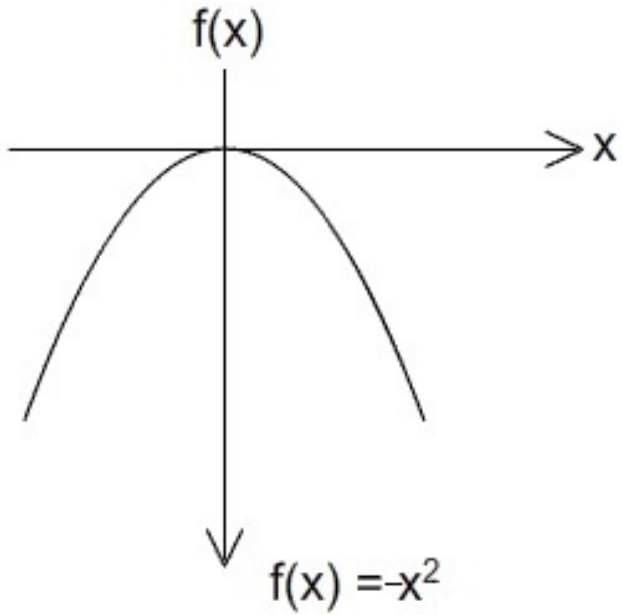
- A function may be increasing for some values of x and decreasing for other values of x.
- Consider the function: $y = x^2 - 4x + 6$
  - The slope of the curve: $y' = 2x - 4$



Maximum and Minimum Values

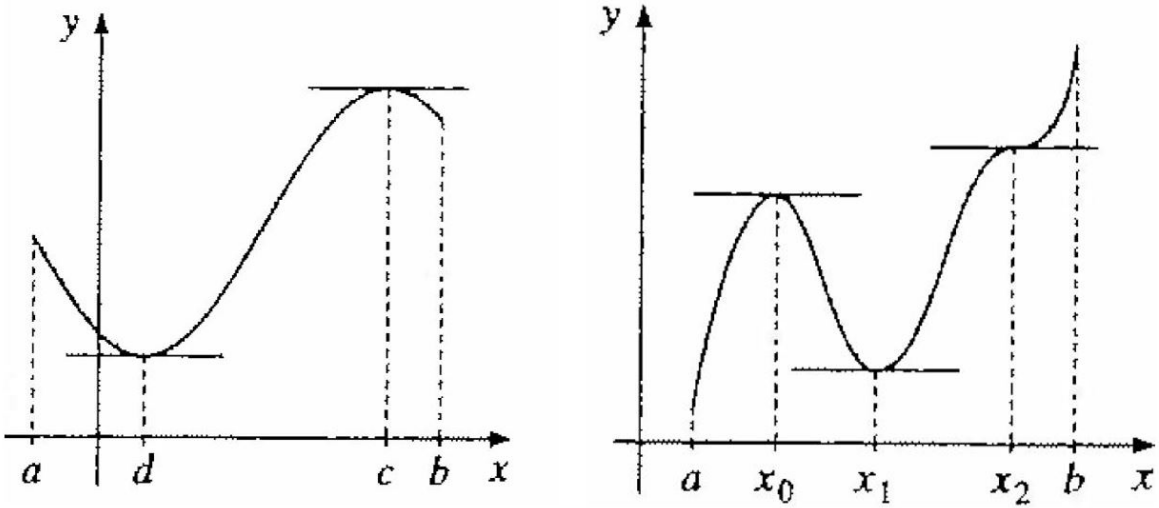
- Definition of **extreme points**:
  - $c \in D$ is a **maximum point** for $f \Rightarrow f(c) \geq f(x)$ for all $x \in D$
  - $d \in D$ is a **minimum point** for $f \Rightarrow f(d) \leq f(x)$ for all $x \in D$
  - Point $x = 0$ is a maximum point for $f(x) = -x^2$ and a minimum point for $f(x) = x^2$



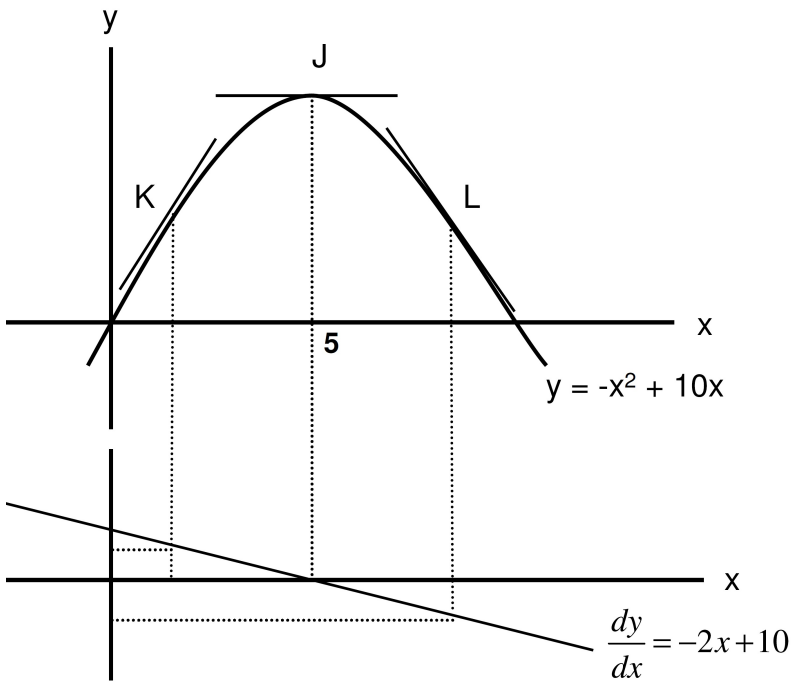


Necessary First-Order Condition for Extreme Points

- The first derivative of a continuous function $y = f(x)$ tells us where that function is increasing, where it is decreasing and where the function reaches a **stationary** point (it is not decreasing or increasing).
  - $f'(c) = 0$ is a necessary condition for c to be a maximum or minimum point.
  - Point x_2 in the right hand graph is called a **point of inflection**.

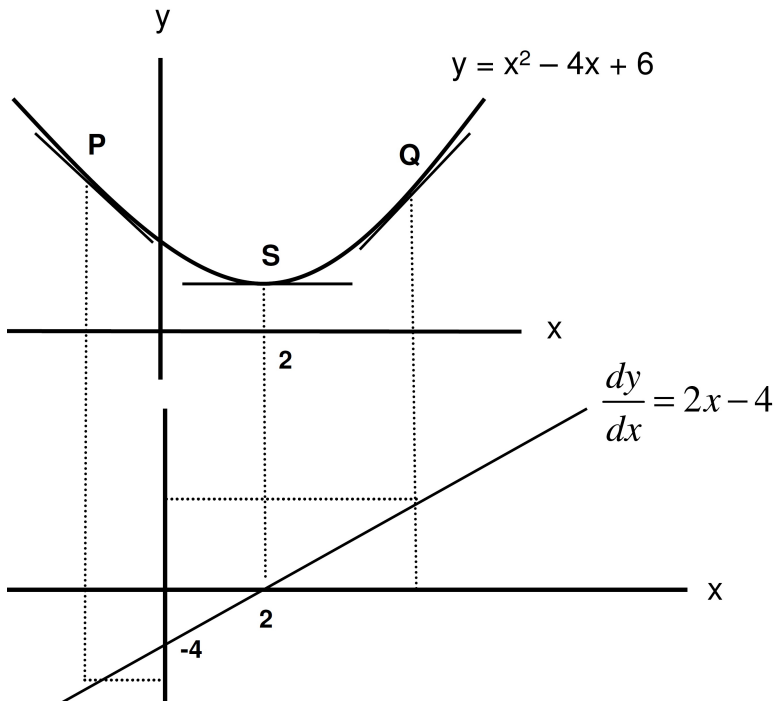


Maximum for Concave Functions



- If $f'(x) \geq 0$ for $x \leq c$, and $f'(x) \leq 0$ for $x \geq c$, then $x = c$ is a maximum point for f.
- Note that the graph of the first derivative is downward-sloping.
- Equivalently, if at point c, $\frac{dy}{dx} = 0$ (the first order condition) and $\frac{d^2y}{dx} < 0$ (the second order condition), the point is a maximum.

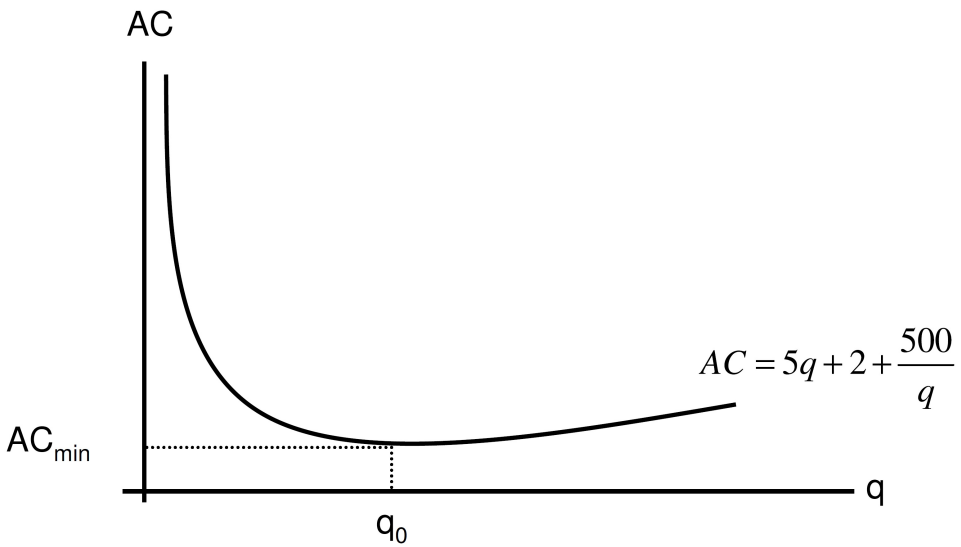
Minimum for Convex Functions



- If $f'(x) \leq 0$ for $x \leq c$, and $f'(x) \geq 0$ for $x \geq c$, then $x = c$ is a minimum point for f.
- Note that the graph of the first derivative is upward-sloping.
- Equivalently, if at point c, $\frac{dy}{dx} = 0$ (the first order condition) and $\frac{d^2y}{dx} > 0$ (the second order condition), the point is a minimum.

Applications: The Firms Cost Function

- The total cost function of a firm can be written as: $TC = f(q) = 5q^2 + 2q + 500$, where q is production units.
  - 500 is the **fixed cost**, and $5q^2 + 2q$ is the **variable cost**
- The **average cost**(AC) $= \dfrac{TC}{q}$



The Minimum Average Cost

- To find the output at which average cost is at its minimum, we take the first order derivative of the average cost function, and make it equal to zero.

$$AC = \frac{TC}{q} = 5q + 2 + 500q^{-1}$$

$$\frac{dAC}{dq} = 5 + 0 + (-1)500q^{-2} = 5 - 500q^{-2} = 0$$

$$5q^2 = 500 \Rightarrow q = \pm 10$$

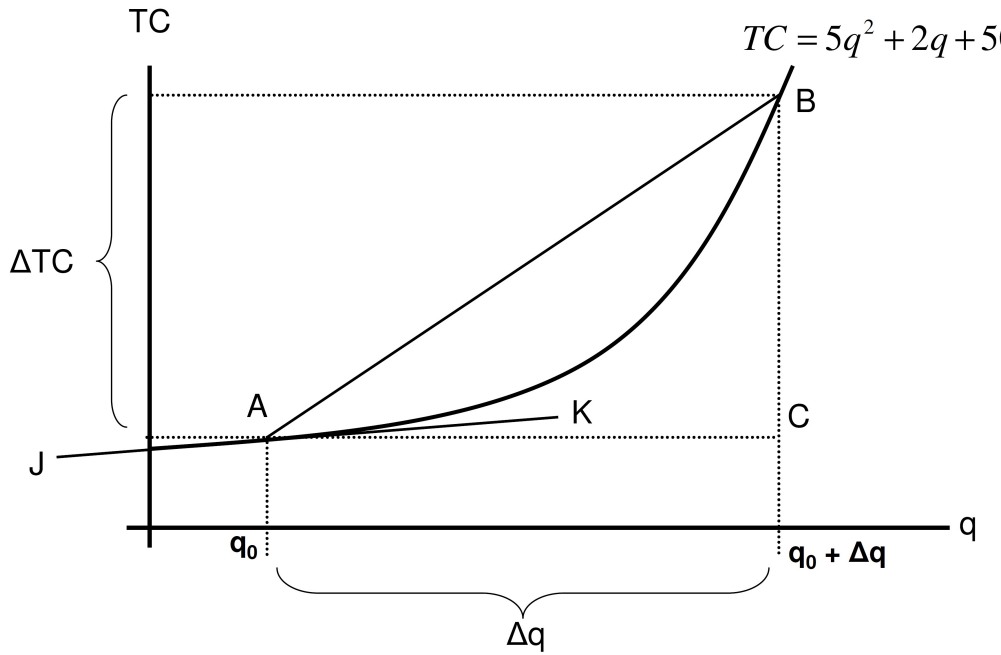
- We ignore the solution $q = -10$, because a negative output does not make sense.
- To check that $q = 10$ is indeed a minimum of the AC function, we take the second derivative and examine its sign at $q = 10$:

$$\frac{d^2AC}{d^2q} = 0 - 500(-2)q^{-3} = \frac{1000}{q^3}$$

$$\left.\frac{d^2AC}{d^2q}\right|_{q=10} = \frac{1000}{10^3} = 1 > 0 \Rightarrow \textit{a minimum at } q = 10$$

Marginal Cost Function

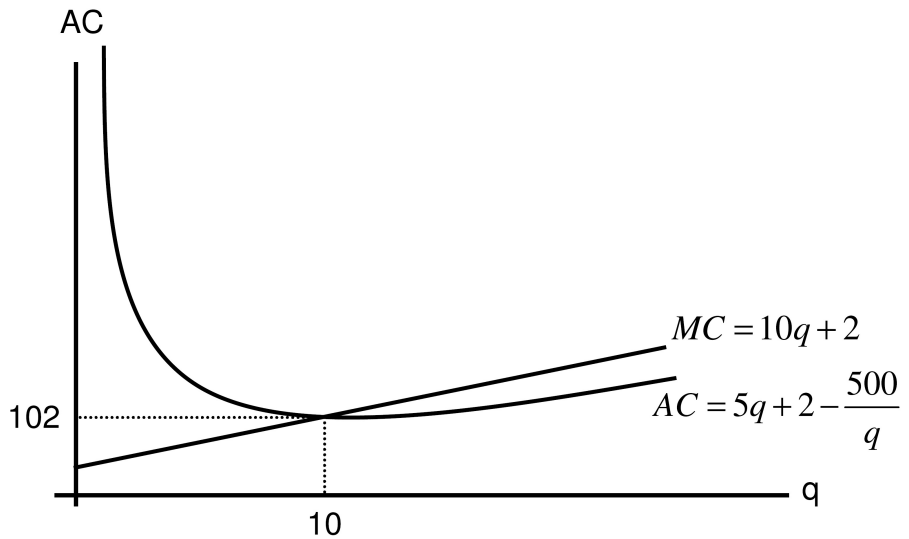
- If we increase the production by a small amount Δq, what is the additional cost of producing those Δq units of good?
- The answer is given by the **Marginal Cost** function (MC): $\dfrac{\Delta TC}{\Delta q}$



- In the limit, $\lim_{\Delta q \to 0} \dfrac{\Delta TC}{\Delta q} = \dfrac{dTC}{dq}$

The Relationship between Average Cost and Marginal Cost

- For the total cost function $TC = 5q^2 + 2q + 500$, the Marginal Cost function is given by:
  - $MC = \dfrac{dTC}{dq} = 10q + 2$
- The Marginal Cost curve measures the slope of the Total Cost curve. The upward sloping Marginal Cost curve reflects the fact that the slope of the Total Cost curve increases as q increases.
- Note that the fixed costs do not affect the Marginal Cost curve.



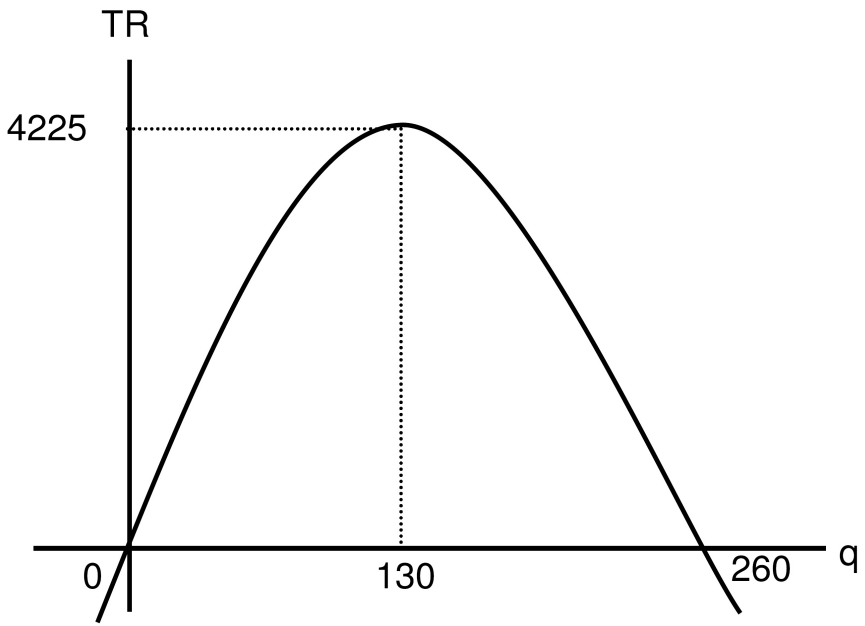
The Relationship between Average Cost and Marginal Cost

- We know that the minimum of Average Cost is where $q = 10$, and hence Average Cost$= 102$.
- To the right of this point (i.e. when $q > 10$), we see that the Marginal Cost curve lies above the Average Cost curve (i.e. Marginal Cost $>$ Average Cost).
- To the left of this point (i.e. when $q < 10$), we see that the Marginal Cost curve lies below the Average Cost curve (i.e. Marginal Cost $<$ Average Cost).
- When $MC = AC$, Average Cost is at its minimum.

$$\begin{aligned} 10q + 2 &= 5q + 2 + 500q^{-1} \\ 10q^2 + 2q &= 5q^2 + 2q + 500 \\ 5q^2 &= 500 \\ q &= \pm 10 \end{aligned}$$

Demand, Total Revenue and Marginal Revenue

- Suppose we are told that the market demand function for a good is given by: $q = -4p + 260$
- Solve for p: $p = -\frac{q}{4} + 65$
- So the total sales revenue (TR) is:
  $TR = pq = (-\frac{q}{4} + 65\ )q = -\frac{q^2}{4} + 65q,$
- Similar to the cost function, we define the marginal revenue function as $MR = \frac{dTR}{dq} = -\frac{q}{2} + 65$



Demand, Total Revenue and Marginal Revenue

- To find the maximum of the Total Revenue function, we take the first derivative, set it equal to zero, and solve for q:

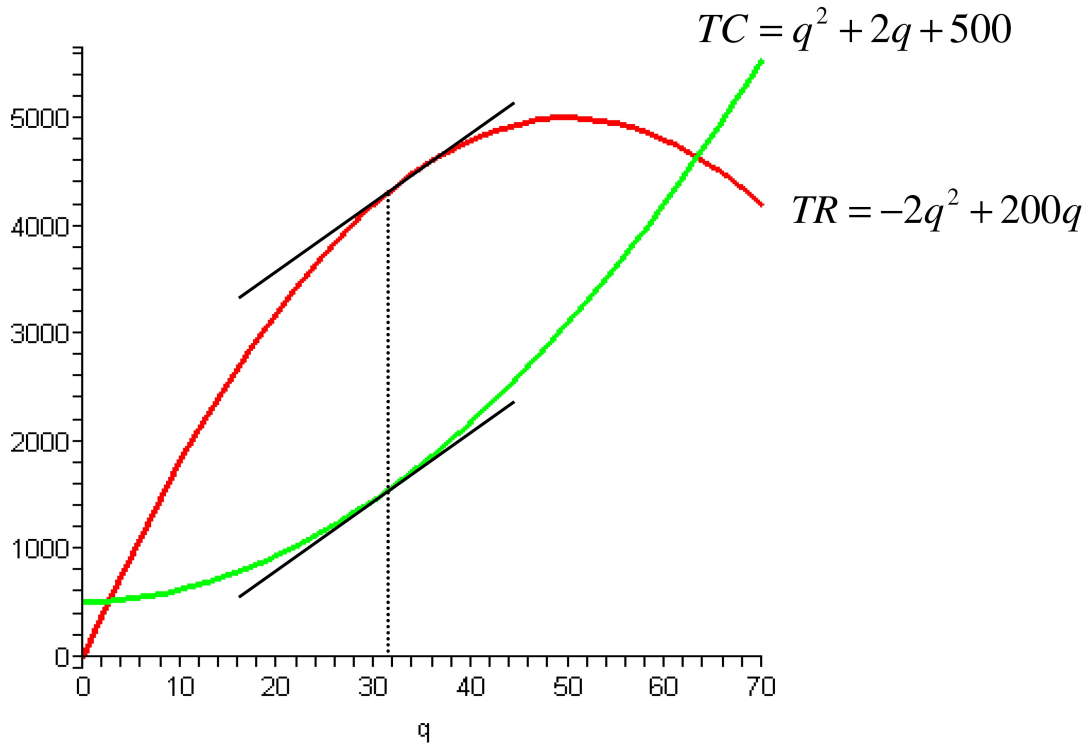
$$\frac{dTR}{dq} = -\frac{2q}{4} + 65 = 0$$
$$\Rightarrow q = 130$$

- To verify that this is a maximum, we take the second derivative and examine its sign:

$$\frac{d^2TR}{d^2q} = -\frac{1}{2} < 0$$

Profit Maximisation

- Given the Total Cost and Total Revenue functions:
  - $TC = q^2 + 2q + 500$, $p = -2q + 200$ and $TR = pq = -2q^2 + 200q$



Profit Maximisation

- Profits equal total revenues from sales minus total production costs:

$$\begin{aligned} \textit{Profits}(\Pi) &= TR - TC \\ \Pi &= -2q^2 + 200q - (q^2 + 2q + 500) \\ \Pi &= -3q^2 + 198q - 500 \end{aligned}$$

- To find the maximum of this function, we take the first derivative, set it equal to zero (first order condition) and solve for q:

$$\frac{d\Pi}{dq} = -6q + 198 = 0 \Rightarrow q = 33$$

- The point $q = 33$ is a maximum if the second derivative is negative:

$$\frac{d^2\Pi}{d^2q} = -6 < 0$$